Class 10 Math - Chapter 5: Arithmetic Progressions

🌟 Introduction

An **Arithmetic Progression (AP)** is a sequence of numbers such that the difference between the consecutive terms is constant. This essentially means every term is obtained by adding a fixed number to the preceding term (except the first term).

The fixed number is called the **common difference** and is denoted by d. It can be positive, negative, or zero!

Standard Form of an AP:

$$ a, a + d, a + 2d, a + 3d, \dots $$

Where:

- $a = \text{first term}$
- $d = \text{common difference}$

📌 Fetching the nth Term (a_n)

If you need to find a specific term deep inside the sequence without writing out the whole list, use the formula for the generalized n-th term:

$$ a_n = a + (n - 1)d $$

> [!NOTE] The general term a_n is sometimes also denoted by l if it represents the **last term** of a finite Arithmetic Progression.

Mind Map of Progression Formulas

```
graph TD;
    AP-->First_Term["First Term (a)"];
    AP-->Common_Difference["Common Difference (d)"];
    AP-->nth_Term["nth Term (a_n)"];
    AP-->Sum_n_Terms["Sum of n terms (S_n)"];
    nth_Term-->Formula1["a_n = a + (n-1)d"];
    Sum_n_Terms-->Formula2["S_n = (n/2)[2a + (n-1)d]"];
    Sum_n_Terms-->Formula3["S_n = (n/2)(a + l)"];
```

⚡ Sum of First n Terms (S_n)

We can add together the first n terms of an AP rapidly using Gauss's mathematical trick summarized in these two formulas:

1. **When given $a, n,$ and d:** $$ S_n = \frac{n}{2} [2a + (n - 1)d] $$
2. **When given just the first term a and last term l:** $$ S_n = \frac{n}{2} (a + l) $$

> [!IMPORTANT] The sum of the first n positive integers ($1 + 2 + 3 + ... + n$) is a fundamental application of this rule: $$ S_n = \frac{n(n + 1)}{2} $$

❌ Common Mistakes

> [!WARNING]
>
> 1. **Calculating d Incorrectly:** The common difference $d = a_2 - a_1$. Always subtract the *previous* term from the *current* term, not the other way around! d can absolutely be negative.
> 2. **Mismatching n and a_n:** n is the *position* index (1st, 2nd, 15th), so it must always be a positive integer. a_n is the actual *value* sitting at that position, which could be an fraction, decimal, or negative number.
> 3. **Using sum formula indiscriminately:** Don't use $S_n = (n/2)(a+l)$ unless you are *certain* the list is finite and you genuinely have the final terminal value l.

📝 Quick Revision

Example Problem

Find the 10th term of the AP: $2, 7, 12, \dots$ **Solution:** Here, $a = 2$. $d = 7 - 2 = 5$. $n = 10$. We use: $a_n = a + (n - 1)d$ $a_{10} = 2 + (10 - 1)5 = 2 + 9(5) = 2 + 45 = 47$. Thus, the 10th term is 47.

Summary

Arithmetic progressions teach us consistency and mathematical linearity. It perfectly models flat-rate interests, seating stadium arrangements, or stacking items consistently. Ensure you distinguish between asking for the *sum* of items versus the *value at a specific spot*.